

Milestone 4 Documentation — DV Unified Military Analytics and Comparison Dashboard

1. Objective

Milestone 4 focuses on final validation, polishing, packaging, and public release of the dashboard system. The goal is to ensure the analytics platform is reliable, accurate, and portfolio-ready for demonstration and evaluation.

Duration: Weeks 7–8

2. Module 7 — Testing and Debugging

This stage validates that every dashboard component works correctly with real data and real user interaction.

2.1 Functional Testing

The following checks were performed across all 4 dashboards:

Area	Test Performed	Expected Result
Filters	Region, continent, alliance filters	Correct dataset subset displayed
Parameters	Country selector (comparison)	Correct country metrics loaded
Navigation	Button switching between pages	Seamless transition without reload issues
Coalition selector	Multi-country aggregation	Combined totals calculated correctly
Tooltips	Hover metrics	Accurate values and labels
KPI cards	Dynamic values	Updates based on filter selection

2.2 Data Validation (Manual QA)

Random country spot checks were performed comparing dashboard values with cleaned dataset.

Validated Metrics: - Defense budget - Active personnel - Aircraft count - Tanks - Naval fleet - Power index rank - Budget-to-GDP ratio

Result: Values matched dataset outputs across sampled countries.

2.3 Performance Testing

- Dashboard load time tested with full dataset (140+ countries)
- Filter switching latency measured
- Coalition aggregation tested with maximum selections

Outcome: - Smooth rendering - No freezing - Acceptable response time

2.4 UI and Layout Fixes

Issues resolved: - Misaligned labels - Incorrect tooltip formatting - Overlapping charts on smaller screens - Navigation button placement - Color contrast improvements for readability

3. Module 8 — Documentation and GitHub Release

3.1 Repository Structure

```
project-root/ | ├── scripts/ | ├── scrape_military_metrics.py | ├── clean_data.ipynb | ├──  
generate_kpis.py | ├── data/ | ├── military_raw_data.csv | ├── military_cleaned.csv | ├──  
military_final.xlsx | ├── dashboard/ | ├── global_military_firepower_2025.pbix / .twbx | ├── docs/  
| ├── project_report.md | ├── milestone_docs.md | └── README.md
```

3.2 README Contents

The README includes: - Project overview - Dataset source description - KPI definitions - Dashboard usage instructions - Installation and execution steps

3.3 How to Use the Dashboard

1. Open the dashboard file (Power BI/Tableau)
 2. Start at Quick Stats page
 3. Apply region or alliance filters
 4. Navigate to Nation Overview for detailed analysis
 5. Use Compare Powers to compare two countries
 6. Use Coalition Builder to simulate alliances
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4. Final Deliverables

- Debugged and polished dashboard workbook
- Organized GitHub repository
- Complete documentation
- Portfolio-ready analytics project
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6. Outcome

Milestone 4 successfully completed the project lifecycle by transforming the developed analytics system into a deployable, shareable, and evaluatable product. The dashboard is now suitable for demonstration, academic evaluation, and portfolio presentation.